

MyCloudX - Personal Cloud Storage System

Problem Statement

Most cloud storage users depend on platforms like Google Drive or iCloud, which come with privacy concerns, recurring fees, and limited control.

MyCloudX addresses this by empowering users to host their own private cloud on a Raspberry Pi – secure, affordable, and fully customizable.

Target User / Audience

- Privacy-focused individuals who demand full control of their data.
- Students, developers, and creators needing flexible, secure storage.
- YouTubers, photographers, and videomakers managing large media collections.
- Families and small teams sharing files safely in a private space.
- DIY and tech enthusiasts who love building self-hosted solutions.

Proposed Solution

MyCloudX transforms a Raspberry Pi into a powerful, personal cloud server.

Users can upload, manage, and share files through an elegant web dashboard, featuring FriendShots – a collaborative photo-sharing space for friends.

No subscriptions, no data mining – just privacy, simplicity, and control.

What Makes MyCloudX Unique

- 100 percent self-hosted cloud – total data ownership.
- Built on Raspberry Pi for affordability, portability, and low power use.
- FriendShots: a unique social feature for group media sharing.
- Modular architecture for scalability and customization.

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- No monthly fees a one-time setup for lifetime privacy.

Core Features

1. Personal Cloud Storage Upload, manage, and access files from any device.
2. FriendShots Collaboration Shared folders for group photos and videos.
3. Secure Authentication Encrypted logins, access control, and user privacy.
4. Scalable Storage Expand easily with SSDs, HDDs, or network NAS.
5. Cloudflare Tunnel Integration Remote, secure access anywhere in the world.

FriendShots Feature

FriendShots creates a private shared album where friends can contribute photos and videos. It brings a social touch to personal cloud storage private, real-time, and free from social media tracking or third-party access.

System Architecture

MyCloudX follows a modular design:

- Frontend: Next.js / React-based dashboard for modern UI.
- Backend: Node.js + Express for file management and user authentication.
- Database: PostgreSQL for metadata, user sessions, and sharing logic.
- Storage: Local SSD or NAS drives mounted on Raspberry Pi.
- Security: Cloudflare Tunnel or HTTPS for encrypted remote access.

Raspberry Pi Hardware Setup

Raspberry Pi 5 (8GB) acts as the main server. It connects to SSD/NVMe storage, Ethernet for stability, and runs on Raspberry Pi OS Lite or Ubuntu Server.

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This setup ensures 24/7 uptime with low energy use.

Technical Specifications

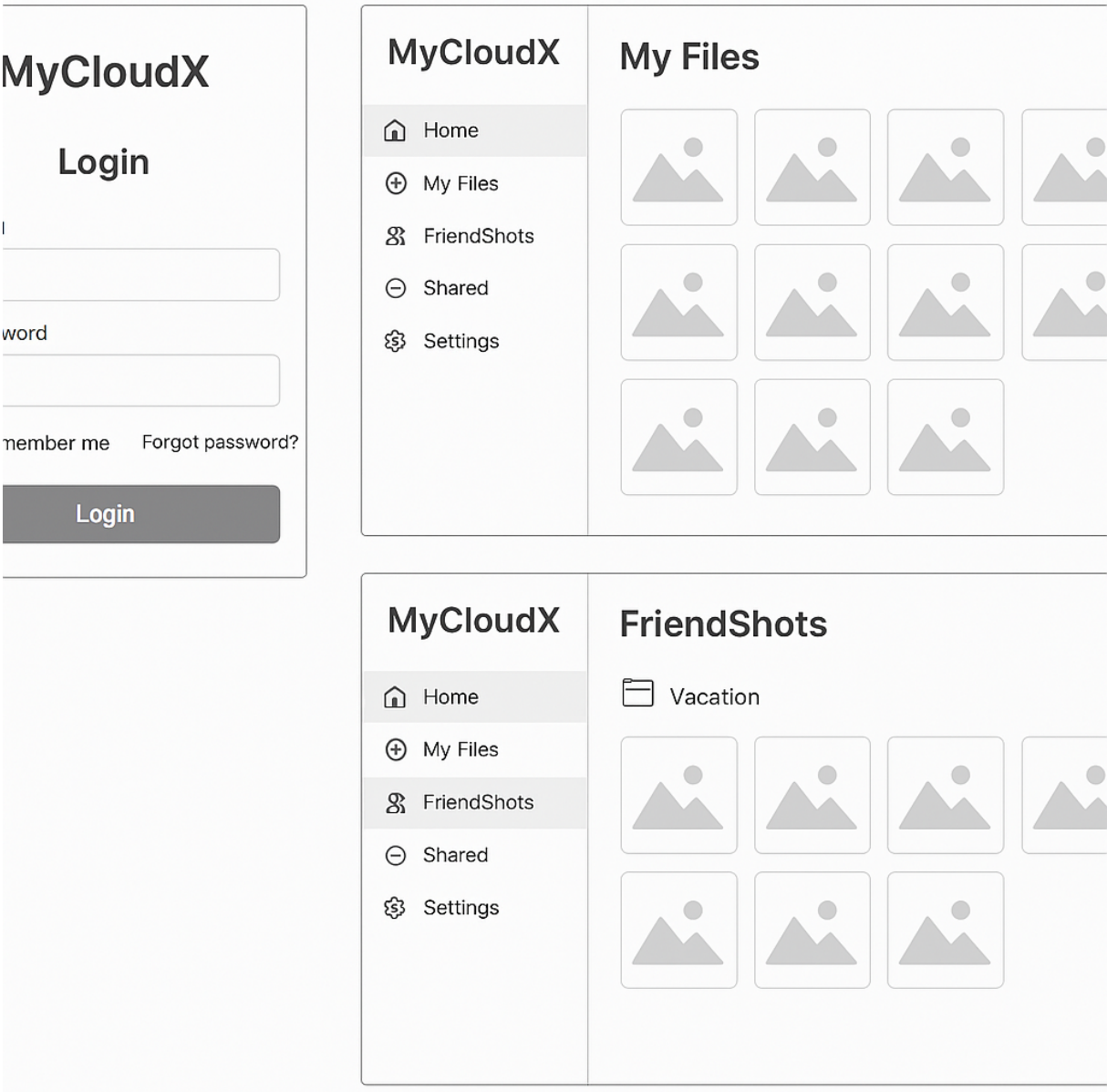
- Raspberry Pi 5 (8GB RAM, 2.4GHz Quad-core CPU)
- 1TB/2TB USB 3.0 SSD or NVMe via PCIe adapter
- PostgreSQL + Node.js + Next.js stack
- Dockerized deployment for easy updates
- HTTPS encryption via Cloudflare Tunnel

Vision & Future Scope

MyCloudX envisions a world where individuals own their digital space.

In the future, AI-driven photo sorting, automatic backups, and mobile syncing will make it as powerful as commercial cloud services but fully under your control.

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Above: Early wireframes for MyCloudX showing login screen, dashboard, and FriendShots view.

[Image Placeholder: Raspberry Pi 5 with SSD and NAS setup]